

Data Integration

□ Data integration

- Combining data from multiple sources into a coherent store

□ Why data integration?

- Help reduce/avoid noise
- Get a more complete picture
- Improve mining speed and quality

□ Schema integration:

- e.g., $A.cust-id \equiv B.cust-\#$
- Integrate metadata from different sources

□ Entity identification:

- Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton

Handling Noise in Data Integration

- ❑ **Detecting data value conflicts**

- ❑ For the same real-world entity, attribute values from different sources are different
- ❑ Possible reasons: no reason, different representations, different scales, e.g., metric vs. British units

- ❑ **Resolving conflict information**

- ❑ Take the mean/median/mode/max/min
- ❑ Take the most recent
- ❑ Truth finding: consider the source quality

- ❑ **Data cleaning + data integration**

Handling Redundancy in Data Integration

- ❑ Redundant data occur often when integration of multiple databases
 - ❑ **Object identification:** The same attribute or object may have different names in different databases
 - ❑ **Derivable data:** One attribute may be a “derived” attribute in another table, e.g., annual revenue
- ❑ What’s the problem?
 - ❑ $Y = 2X \rightarrow Y = X_1 + X_2 \quad Y = 3X_1 - X_2 \quad Y = -1291X_1 + 1293X_2$
- ❑ Redundant attributes may be detected by correlation analysis and covariance analysis